

1. Ispitajte djeljivost izraza, odredite kvocijent i ostatak dijeljenja:

a) $4x^4 + 3x^2 - x + 2$ i $x + 2$

b) $x^5 + x^4 - 2x + 4$ i $x^3 - 2x + 2$

2. Odredite koeficijente polinoma čija su rješenja : 3 , 2, 1

3. Izračunajte vrijednost polinoma $p(x) = -2x^4 + 3x^2 + 1$ za $x = -1$

4. Zadan je univerzalni skup $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$, a skupovi $A = \{1, 2, 3, 5, 7, 8, 9\}$ i $B = \{1, 2, 3, 4, 6, 8, 12\}$, $C = \{10, 11, 12\}$ njegovi su podskupovi. Odrediti :

$$A^c \cup B^c, A^c \cap B^c, A \Delta B, (A \setminus C)^c \cap B^c.$$

5. Ispitajte jeli matrice $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ singularna? Odredite broj redaka i stupaca, izdvojite

drugi stupac, treći redak , element a_{31}

6. Riješite matricnu jednadžbu $A \cdot X + 2B = C + B \cdot X$ ako je

$$A = \begin{bmatrix} 3 & -1 \\ 5 & -2 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \text{ i } C = \begin{bmatrix} 14 & 16 \\ 9 & 10 \end{bmatrix}.$$

7. Riješite sustav linearnih jednadžbi:

$$\begin{aligned} x + y + 5z + 2u &= 1 \\ 2x + 3y + 11z + 5u &= 2 \\ 2x + y + 3z + 2u &= -3 \\ x + y + 3z + 3u &= -3 \end{aligned}$$

8. Definirati varijablu x kao $x=1.5$ i izračunati: $\frac{\sqrt{80x^3}}{e^{3x}}$.

9. Izračunati vrijednosti izraza :

a) $\log 25 + \ln e^4$ b) $\sqrt{\log 3^2 + 1}$,

10. Zadani su radij vektori $\vec{r}_A = (1,1,1)$, $\vec{r}_B = (4,4,4)$, $\vec{r}_C = (3,5,5)$ i $\vec{r}_D = (2,2,3)$. Odredite volumen paralelopipeda kojeg razapinju $\vec{r}_A, \vec{r}_C$ i $\vec{r}_D$. Odredite vektorsku projekciju vektora $\vec{CD}$ na vektor $\vec{BA}$

11. Vrhovi trokuta su $A(1,1,1), B(0,2,2), C(2,1,3)$. Odredite:

a) kut u vrhu A b) opseg trokuta c) skalarni produkt vectora $\vec{AB}$ i $\vec{BC}$.

12. Na istom crtežu usporedite grafove funkcija $\sin 16\pi x$ i $\cos 16\pi x$.

13. Grafički prikažite nul-točke polinoma i provjerite rješenje (koristeći naredbu roots)

$$P(k) = k^3 + k^2 - 37k + 35$$

14. Grafički prikazite slijedeće funkcije : $y = \frac{(x-1)^2}{-x^2+2x+8}$

15. Izračunati:

(a) $\operatorname{Re} \frac{1-i}{(2+i)(3-i)}$; (b) $\operatorname{Im} \frac{(2-i)^2}{(2+i)(3+i)}$; (c) $\operatorname{Re} \frac{i}{(2+i)^2}$;

(d) $\operatorname{Im} \frac{(1-i)^3}{-2i}$; (e) $\operatorname{Re} \frac{\overline{5-i}}{i(2+i)}$; (f) $\operatorname{Im} \frac{(2+i)i^2}{i-5}$;

(g) $\operatorname{Re} \frac{(1-2i)^2}{(3-i)(3+i)}$; (h) $\operatorname{Im} \frac{2(5-i)^2}{(2+i)(3-i)}$; (i) $\operatorname{Re} \frac{i^{123} - i^{57}}{i^{11} + i^{303}}$.

Rješenja:

1.a)

$p=[4,0,3,-1,2]; q=[1,2];$

$\gg [d,R]=\operatorname{deconv}(p,q)$

$d =$

4 -8 19 -39

$R =$

0 0 0 0 80

2.

$\gg r=[3,2,1]; \operatorname{poly}(r)$

$\operatorname{ans} =$

1 -6 11 -6

3.

```
>> p=[-2,0,3,0,1]
```

```
p =
```

```
-2    0    3    0    1
```

```
>> polyval(p,-1)
```

```
ans =
```

```
2
```

4.

```
>> U=1:12;A=[1,2,3,5,7,8,9];B=[1,2,3,4,6,8,12];C=[10,11,12];
```

```
>> union(setdiff(U,A),setdiff(U,B))
```

```
ans =
```

```
4    5    6    7    9   10   11   12
```

```
>> intersect(setdiff(U,A),setdiff(U,B))
```

```
ans =
```

```
10   11
```

```
>> union(setdiff(A,B),setdiff(B,A))
```

```
ans =
```

```
4    5    6    7    9   12
```

```
>> intersect(setdiff(U,setdiff(A,C)),setdiff(U,B))
```

```
ans =
```

```
10   11
```

5.

```
>> A=[1,2,1;0,1,1;1,2,2]
```

```
A =
```

```
1 2 1
0 1 1
1 2 2
```

```
>> det(A)
```

```
ans =
```

```
1
```

```
>> size(A,1),size(A,2),A(:,2),A(3,:),A(3,1)
```

```
ans =
```

```
3
```

```
ans =
```

```
3
```

```
ans =
```

```
2
```

```
1
```

```
2
```

```
ans =
```

```
1 2 2
```

```
ans =
```

```
1
```

6.

```
>> A=[3,-1;5,-2];B=[5,6;7,8];C=[14,16;9,10];
```

```
>> X=inv(A-B)*(C-2*B)
```

```
X =
```

```
-12.500000000000000 -13.666666666666666
3.000000000000000 3.333333333333333
```

```
>> format rational
>> X=inv(A-B)*(C-2*B)
```

X =

```
-25/2    -41/3
      3    10/3
```

7.

```
>> A=[1,1,5,2;2,3,11,5;2,1,3,2;1,1,3,3];b=[1;2;-3;-3];
>> X=A\b
```

X =

```
-7/4
 5/8
 9/8
-7/4
```

```
>> rref([A,b])
```

ans =

```
1      0      0      0    -7/4
0      1      0      0     5/8
0      0      1      0     9/8
0      0      0      1    -7/4
```

8.

```
>> format short
>> x=1.5;(sqrt(80*x^3))/(exp(3*x))
```

ans =

```
0.1825
```

9.

```
>> log10(25)+log(exp(4))
```

ans =

```
5.3979
```

```
>> sqrt(log10(3^2)+1)
```

```
ans =
```

```
1.3979
```

10.

```
>> ra=[1,1,1];rb=[4,4,4];rc=[3,5,5];rd=[2,2,3];
```

```
>> v=det([ra;rc;rd])
```

```
v =
```

```
2
```

```
>> cd=rd-rc
```

```
cd =
```

```
-1 -3 -2
```

```
>> ba=ra-rb
```

```
ba =
```

```
-3 -3 -3
```

```
>> proj=(dot(cd,ba)/dot(ba,ba))*ba
```

```
proj =
```

```
-2 -2 -2
```

11.

```
>> A=[1,1,1];B=[0,2,2];C=[2,1,3];
```

```
>> ab=B-A
```

```
ab =
```

```
-1 1 1
```

```
>> ac=C-A
```

```
ac =
```

1 0 2

```
>> bc=C-B
```

bc =

2 -1 1

```
alfa=acos((dot(ab,ac))/(norm(ab)*norm(ac)))
```

alfa =

1.3096

```
>> opseg=norm(ab)+norm(bc)+norm(ac)
```

opseg =

6.4176

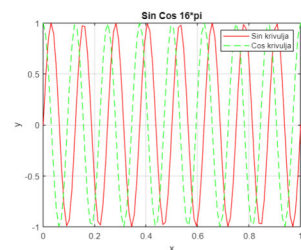
```
>> dot(ab,bc)
```

ans =

-2

12.

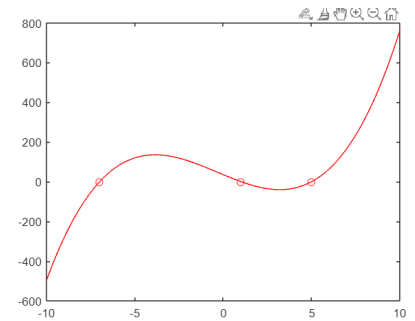
```
>> x = linspace (0,1,100);  
>> plot(x, sin(16*pi*x),'r-',x,cos(16*pi*x),'g--');  
>> legend('Sin krivulja','Cos krivulja');  
title('Sin Cos 16*pi');  
>> xlabel('x'), ylabel('y')  
>> grid
```



13.

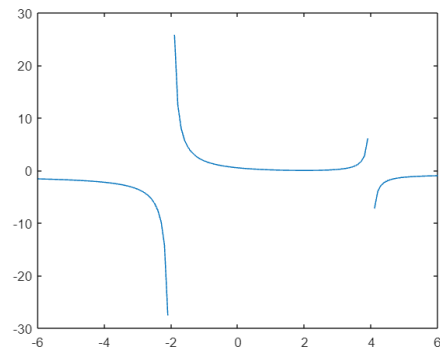
$$P(k) = k^3 + k^2 - 37k + 35$$

```
x=-10:0.1:10;
f=[1 1 -37 35];
plot(x,polyval(f,x), 'r-', roots(f),0, 'ro')
```



14.

```
x=-6:0.1:6;
y=(x-2).^2./(-x.^2+2*x+8);
plot(x,y)
```



15.

format rational

a) $\frac{3}{25}$; (b) $-\frac{7}{10}$; (c) $\frac{4}{25}$; (d) -1 ; (e) $-\frac{3}{5}$; (f) $\frac{3}{26}$; (g) $-\frac{3}{10}$; (h) $\frac{14}{5}$; (i) 1